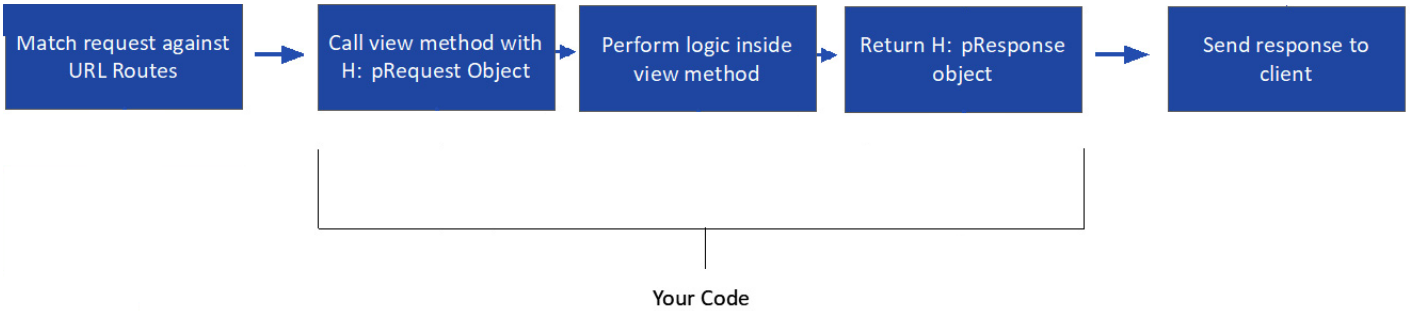
This lab session covers the basics of the Django URL and HTML template. By the end of this session, students should have a very good understanding of how to define URL patterns and create aan HTML template and how both work together to Processing an HTTP request. They should also have already practiced with the Git system and how to use basic commands to push and manage their repository.

The following diagram shows the direction of the transmission of HTTP requests and HTTP responses, between a browser and a web server

Inside the framework, the flow of HTTP request and response is illustrated in the following figure. The sections indicated as **Your Code** are for the code that you write, and the first and last steps are taken care of by Django. Django does the URL matching for you, calls your view code, and then handles passing the response back to the client. Actually, you are going to build and configure a link between a URL and a view where the URL matching is taking place by Django.



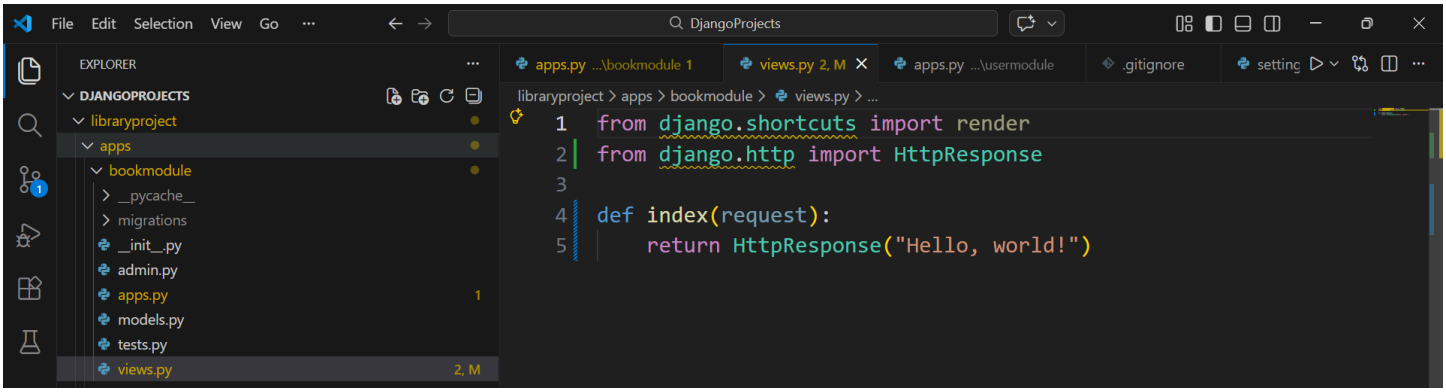
Pre-lab Preparation:

1. Complete the previous lab (Lab3): must have the initial Django project with apps (bookmodule & usermodule) and the necessaries configurations.
2. Good understanding of HTTP protocol, mainly the request, response, and HTTP status codes.
3. Basic knowledge about HTTP tags.

Lab Activities: Build and configure a simple link between a URL and a view (simple pattern), along with a simple HTML template.

Task 1: Build your first view function and corresponding URL mapping in the core/urls.py

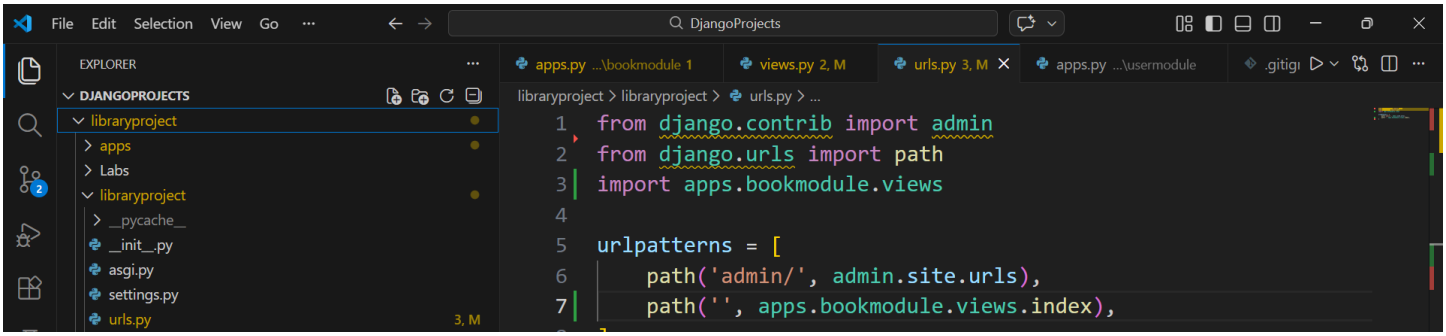
Step 1:



Step 2:

```
import apps.bookmodule.views
```

Step 3:



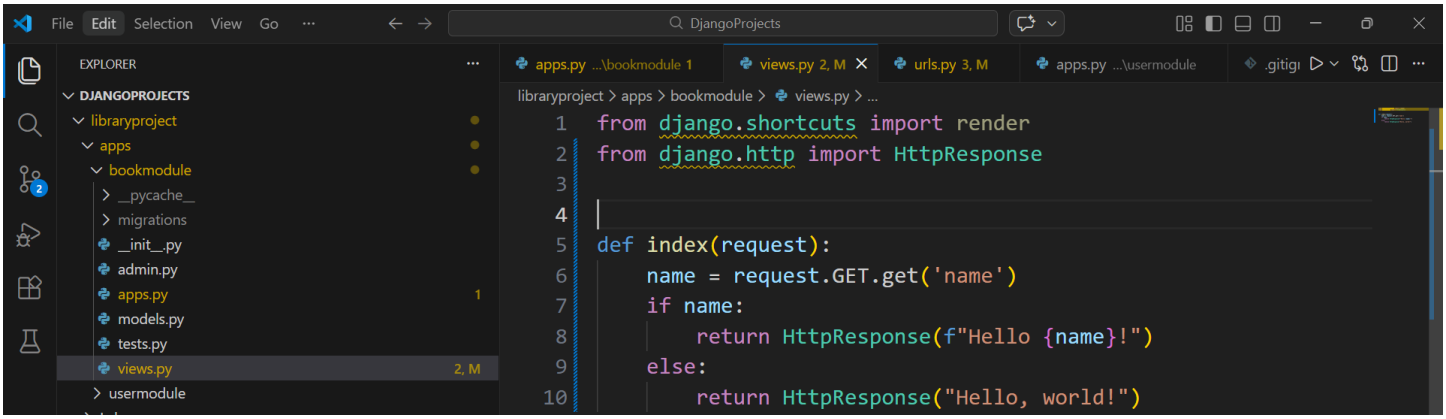
Step 4:

```
(djangoenv) PS C:\Users\re2na\OneDrive\طس\بتكمل\T_W\practice\DjangoProjects\libraryproject> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).

You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, conten
ttypes, sessions.
Run 'python manage.py migrate' to apply them.
March 03, 2026 - 02:26:51
Django version 5.2.11, using settings 'libraryproject.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/
```

Step 5: Open a web browser and navigate to http://127.0.0.1:8000. You should see "`Hello, world!`"

Task 2: Add parameters with HTTP requests[1]

Step 1:

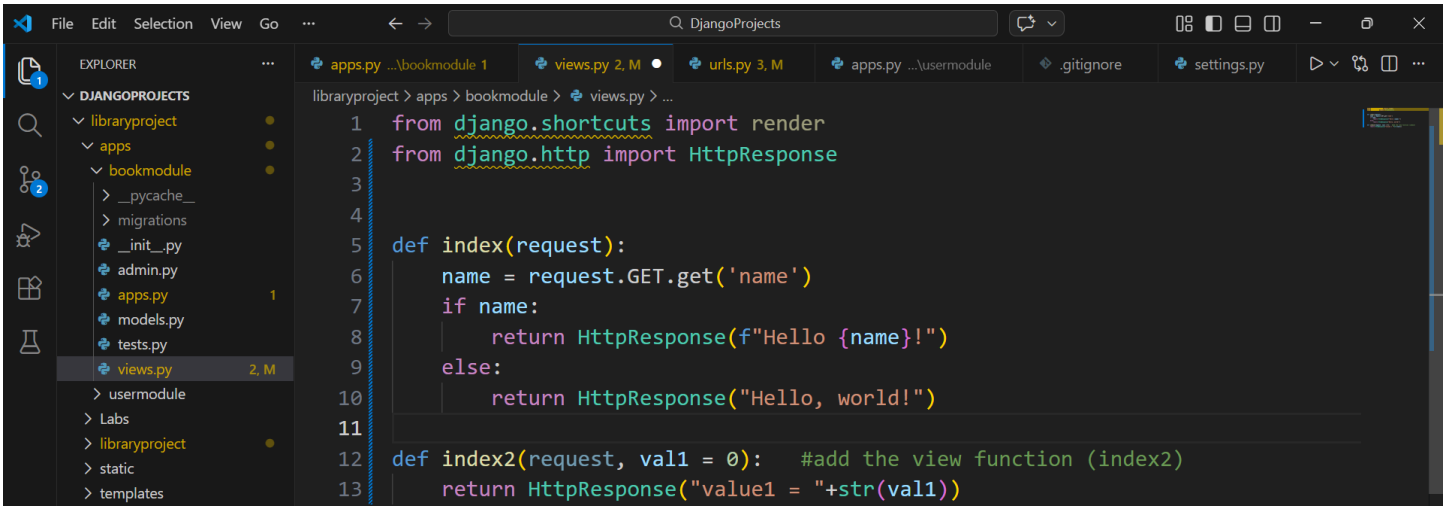


Step 2: Now from the browser navigate to the following urls:

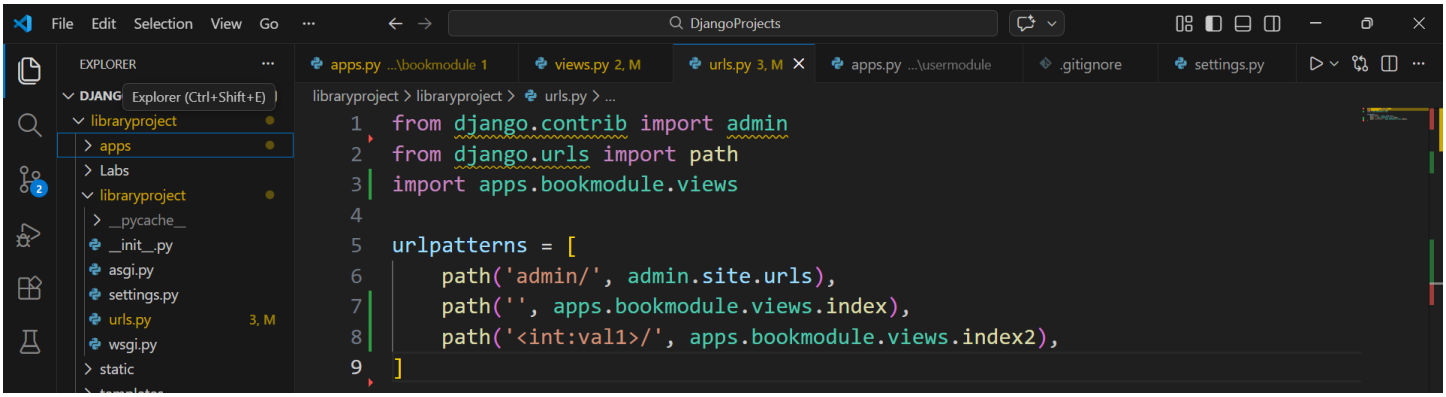
[1] It is possible to have more than one parameter, like http://127.0.0.1:8000?id=10&name=world!&age=18
Note: this method is GET HTTP request, which is not recommended for security reasons, and use POST HTTP request instead.

Task 3: Build your second view function and corresponding URL mapping with parameters within URL path

Step 1:



Step 2:



Step 3:

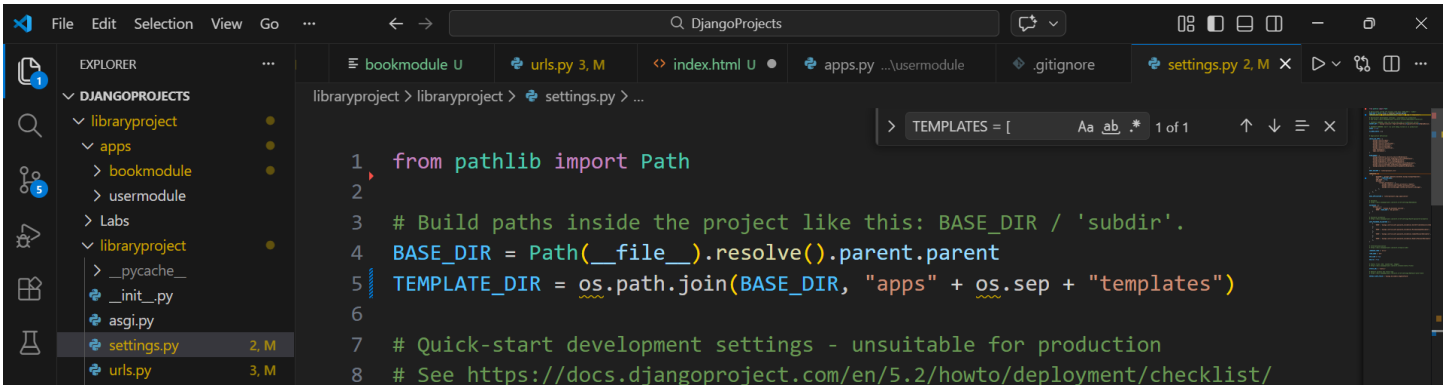
value1 = 3

Task 4: Create a simple HTML template

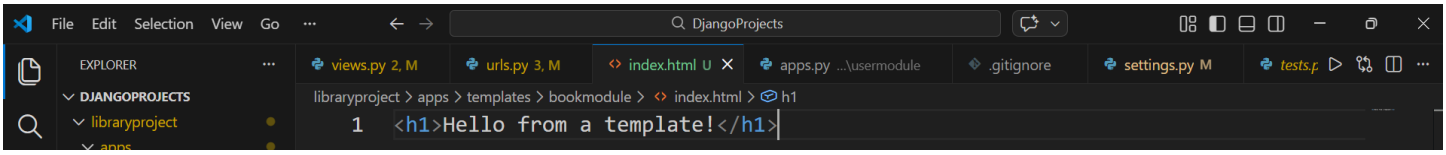
Step 1: Create a templates directory for bookmodule app, by doing the following:

```
> mkdir apps/templates/bookmodule
```

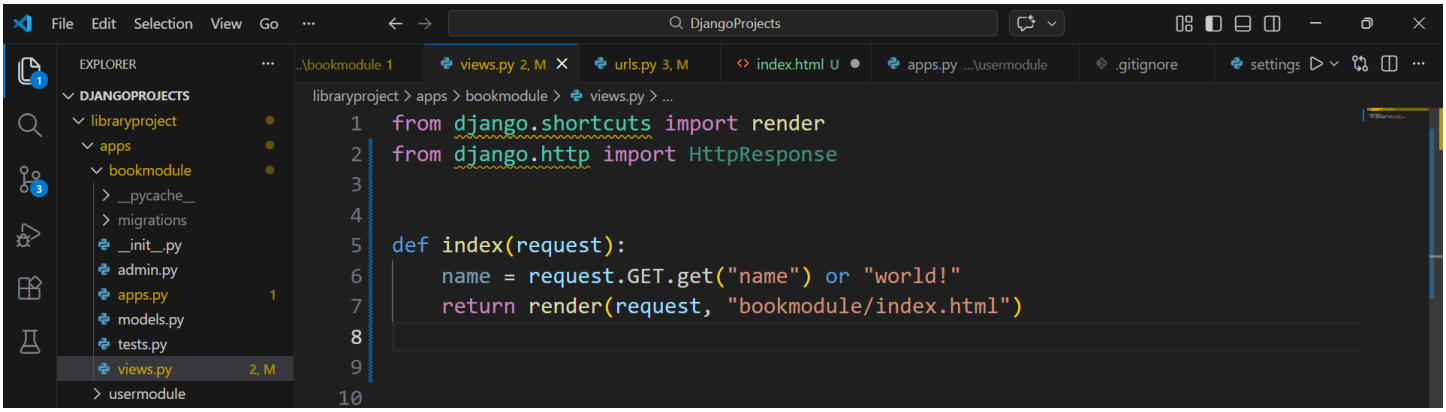
Step 2: Update main/settings.py with template location



Step 3:



Step 4: Step 5: Update the index function, in apps/bookmodule/views.py, so that, instead of returning HttpResponse, it's returning a call to render, passing in request and the template name:

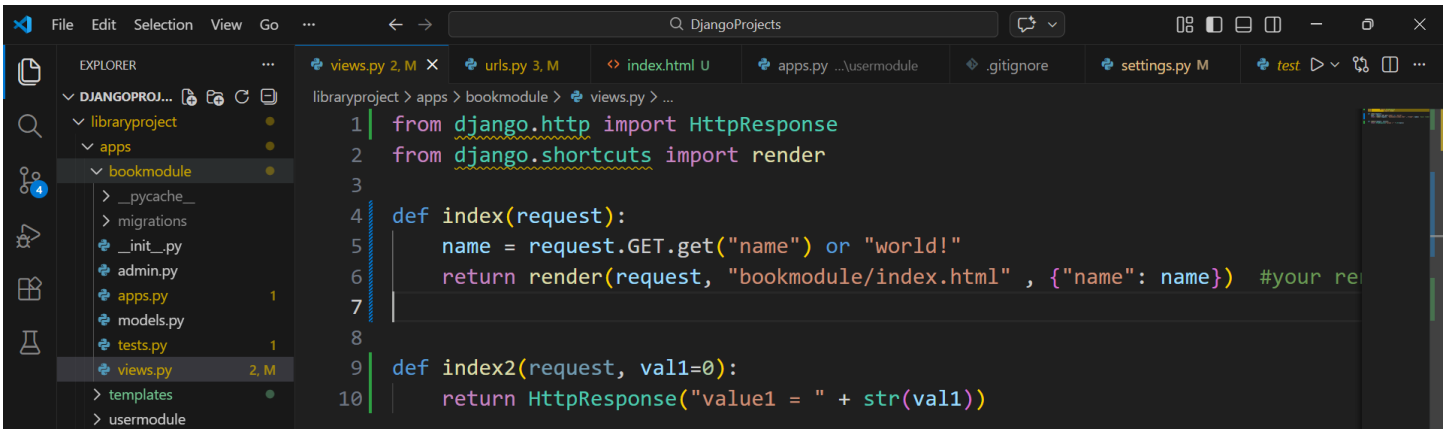


Hello from a template!

Task 5: Rendering variables in the HTML template that processes a context

Step 1:

Step 2:



Task 6: Define patterns globally (DjangoProjects/urls.py) with specific urls.py file for each app/module

Note that it is common to keep one URL configuration per application in your Django project. Here, we will create a separate URL configuration for each app and add it to our project-level URL configuration.

Step 1: Create a new urls.py file for each app module(bookmodule and usermodule) in the same folder as the views.py file.

Step 2: move urls of bookmodule to apps/bookmodule/urls.py, and include newly created urls.py in core/urls.py using include module.

DjangoProjects/urls.py

```
from django.contrib import admin
from django.urls import include, path
urlpatterns = [
    path('admin/', admin.site.urls),
    path('books/', include("apps.bookmodule.urls")), #include urls.py of bookmodule app
    path('users/', include("apps.usermodule.urls"))  #include urls.py of usermodule app
]
```

apps/bookmodule/urls.py

```
from django.urls import path
from . import views
```

```
urlpatterns = [
    path('', views.index),
    path('index2/<int:val1>/', views.index2)
]
```

apps/usermodule/urls.py

```
from django.urls import path
urlpatterns = [
]
```

Step 3:

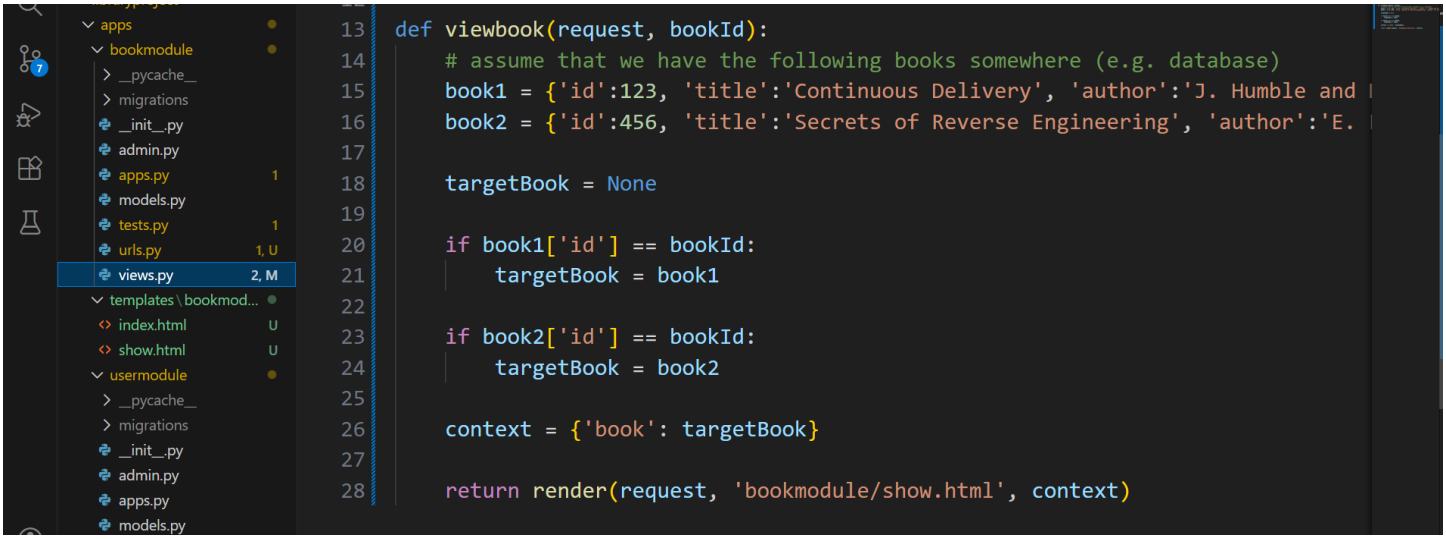
Hello from world!!

Hello from world!!

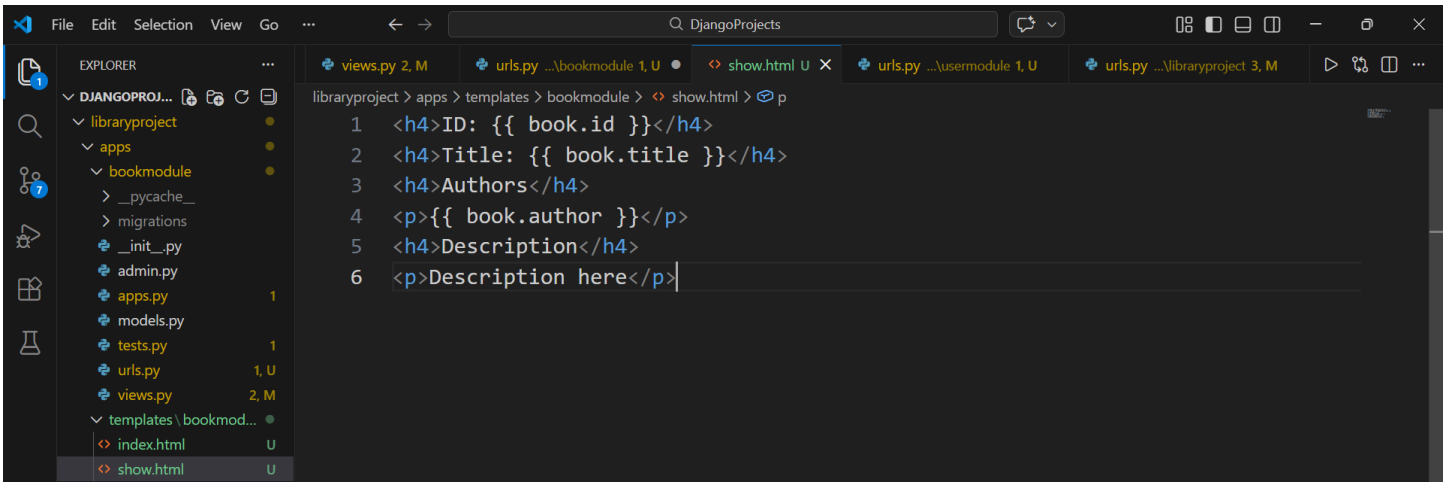
value1 = 3

Task7: Create a URL, view, and HTML to display one book details

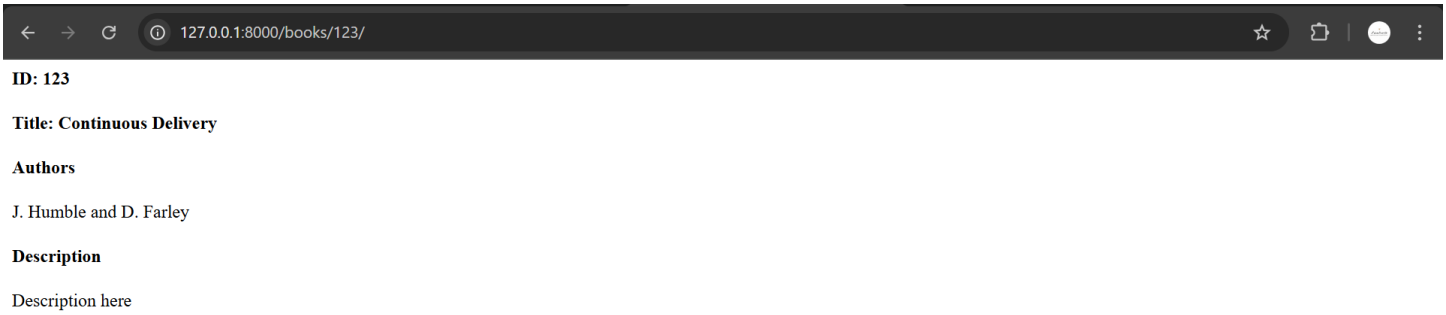
Step 1:



Step 3:



Step 4:



Additional content: To redirect to another view or URL, use the function redirect, and for details and examples visit the following link:
https://docs.djangoproject.com/en/5.1/topics/http/shortcuts/#redirect